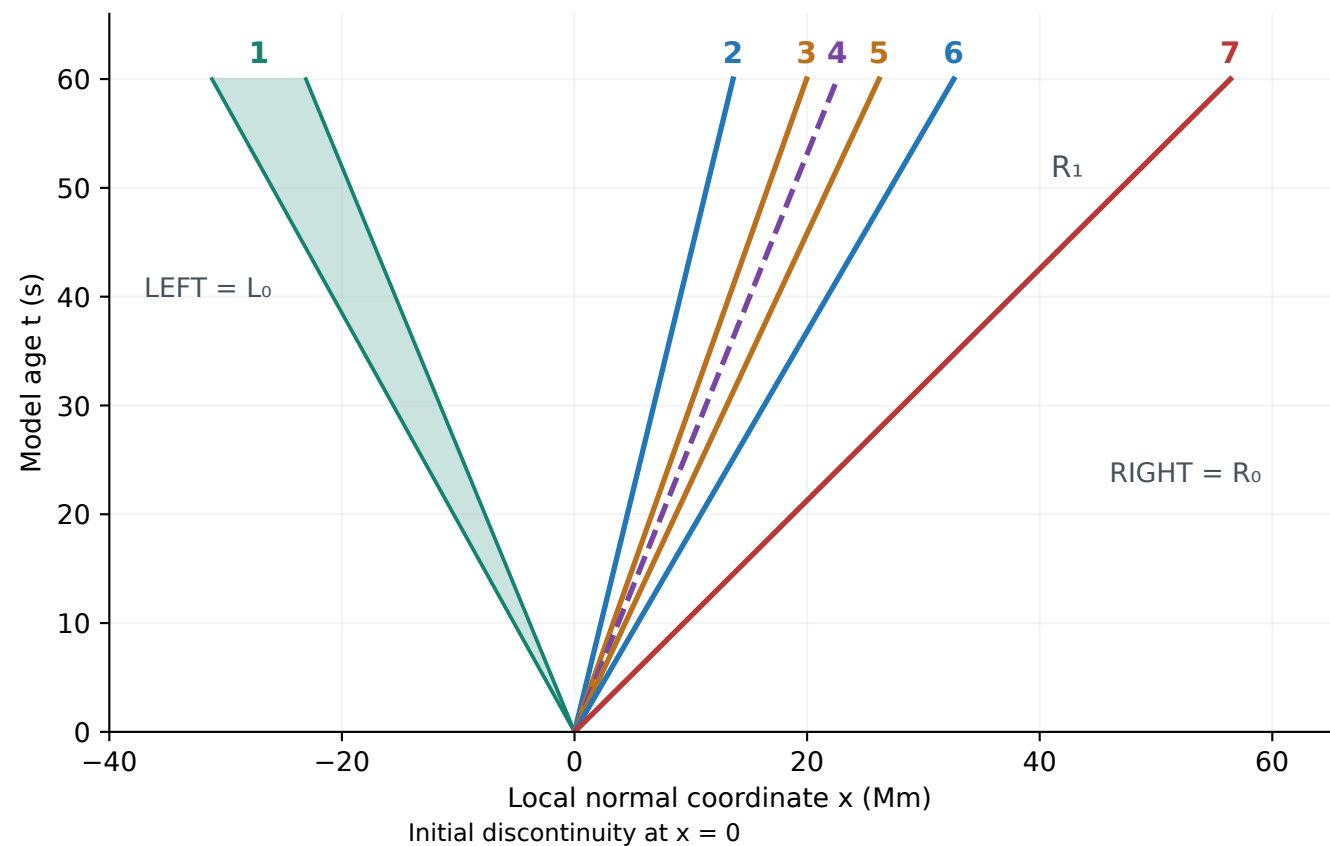


An MHD Riemann Fan for a Solar EUV Front

Two specified plasma states evolve into a connected set of waves.

Computed ideal-MHD example: assumed local states for the 10 September 2017 context.



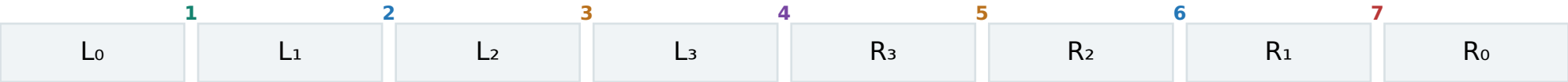
THE SEVEN WAVES

- 1** Fast rarefaction
-519.5 to -384.9 km/s
- 2** Alfvén / rotation
227.3 km/s
- 3** Slow shock
332.6 km/s
- 4** Contact
376.5 km/s
- 5** Slow shock
436.5 km/s
- 6** Alfvén / rotation
543.5 km/s
- 7** Fast shock
940.0 km/s

Speeds are in the ambient rest frame.
Model age is not an observed launch time.

EIGHT CONNECTED STATES

Order only; boxes are not spatial widths



Initial L_0 / R_0 define the whole fan. Across the outer fast shock, test the adjacent pair R_1 / R_0 .

MODEL FEATURE	PHYSICAL CHECK	OBSERVATIONAL TARGET
1 Fast rarefaction	Continuous expansion; constant p / ρ^γ	Density, temperature, flow and field profiles
2, 6 Alfvén waves	$\rho, p, u_n, B_t $ fixed; vector rotation + Walén test	Transverse velocity and magnetic-field changes
3, 5 Slow shocks	Conservation + entropy + slow characteristics	Adjacent states; especially magnetic geometry
4 Contact	u and B continuous, p equal; density may jump	Density/temperature contrast; material motion
7 Fast shock	Conservation + entropy + fast characteristics	Compression, heating and front-relative flow

EUV + white light: geometry and density constraints. Spectroscopy: thermal / LOS-flow constraints.
Magnetic diagnostics + viewing geometry constrain the vector state. Seven waves need not give seven bright ridges.

All seven amplitudes are nonzero for these inputs; other inputs may remove waves or change shock / rarefaction branches.
Ideal isotropic adiabatic MHD, $\gamma = 5/3$, $B_n \neq 0$. Measured radiance does not yet establish these eight solar plasma states.